Theorem 16: inapproximability of the optimal-transport local edit distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem16_LocalEditHardness`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 16 (Section 4, p. 12, l. 424–425; proof in Appendix G, pp. 51–52, l. 1621–1670): *for any* $\varepsilon > 0$*, unless* P = NP*, there is no polynomial-time approximation algorithm computing* d_1^{OT} *with multiplicative error guarantee* $2-\varepsilon$*, even for graphs of maximum degree two.* It is the case $p = 1$ of Theorem 4 (2) (p. 4, l. 135–137). The reduction maps an instance $(A, a_1, \dots, a_{3m})$ of Three-Partition ($\sum_i a_i = mA$, $A/4 < a_i < A/2$, unary) to F_1, the disjoint union of paths on a_i vertices, and F_2, the disjoint union of m cycles on A vertices, and shows the gap: YES-instances have $d_1^{\mathrm{OT}}(F_1, F_2) \le 1$, NO-instances $d_1^{\mathrm{OT}}(F_1, F_2) \ge 2$ (shown for $d_1^{\bullet}$, l. 1624–1625, and transferred to d_1^{OT} by Theorem 1, l. 1667–1668).

Lean. Only the gap (the correctness of the reduction) is formalised. With $F_1 =$ `pathUnion I.a` and $F_2 =$ `cycleUnion I.m I.A`:

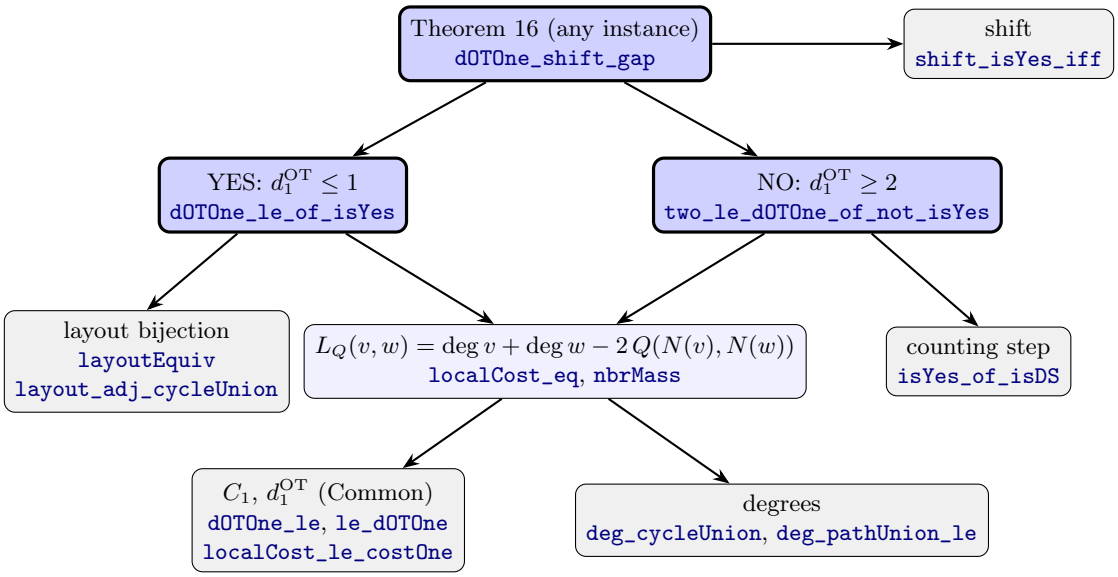
- `GsimOT.Hardness.ThreePartition.dOTOne_le_of_isYes`: if $a_i \ge 2$ for all i and I is a YES-instance, then $d_1^{\mathrm{OT}}(F_1, F_2) \le 1$;
- `GsimOT.Hardness.ThreePartition.two_le_dOTOne_of_not_isYes`: if I is a NO-instance, then $d_1^{\mathrm{OT}}(F_1, F_2) \ge 2$ (no hypothesis on the a_i);
- `GsimOT.Hardness.ThreePartition.dOTOne_shift_gap`: both, for every instance, after the shift of l. 1622–1624 (add 3 to each a_i, 9 to A).

$d_1^{\mathrm{OT}}(G, H) = \inf_{Q \in \mathrm{DS}(V,W)} C_1(Q)$ with $C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v, w)$ is `GsimOT.dOTOne` (`Common/Basic.lean`), the definition of Proposition 8 and Theorem 9; by Eq. (7) it equals the operator-norm distance $d_{1\to 1}^{\mathrm{OT}}$ of the paper (`GsimOT.dOT_one_one_eq_dOTOne`). Both graphs have maximum degree two (`deg_pathUnion_le`, `deg_cycleUnion`).

2 Where the proof lives

File	Content
`Theorem16_LocalEditHardness/Theorem16.lean`	$Q(N(v), N(w))$ (`nbrMass`), the local-cost formula `localCost_eq`, the YES- and NO-directions, the shifted version.
`Common/ThreePartition.lean`	Three-Partition instances, the layout bijection (`layoutEquiv`, `layout_adj_cycleUnion`), the counting step `isYes_of_isDS`, the shift.
`Common/PathsAndCycles.lean`	`pathUnion`, `cycleUnion`, degrees, constancy along paths (`eq_of_pathUnion_adj`).
`Common/Basic.lean`, `Common/MatrixNorms.lean`	`dOTOne`, `costOne`, `localCost`, `dOTOne_le`, `le_dOTOne`, `localCost_le_costOne`, `localCost_eq_sum_col`, Eq. (7).

3 Proof layout



Blue: this folder; grey: `GsimOT/Common`. Nothing is cited. (The YES- and NO-proofs also use the C_1 lemmas and the degrees directly.)

4 The proof step by step

Step 0: w.l.o.g. $a_i \ge 3$ (l. 1622–1624). Add 3 to every a_i and 9 to A: `ThreePartition.shift`, `shift_isYes_iff`; used in `dOTOne_shift_gap`.

Step 1: blowups (l. 1624–1625, l. 1667). The paper proves the bounds for the blowup distance $d_1^{\bullet}$, by looking at permutations of the k-blowups, and transfers them with Theorem 1 ($d_1^{\mathrm{OT}} = d_1^{\bullet}$). *The formalisation skips this step* and proves the bounds for every $Q \in \mathrm{DS}(V,W)$ directly; see Section 6. The key identity is (`localCost_eq`)

$$L_Q(v,w) = \deg_{F_1}(v) + \deg_{F_2}(w) - 2\,Q(N(v),N(w)), \qquad Q(N(v),N(w)) = \sum_{v'\sim v,\ w'\sim w} Q(v',w'),$$

the fractional form of the paper's mismatch count "$(2k-\ell)+(dk-\ell)$" (l. 1661): $\deg(v)$ edges to lose, $\deg(w)=2$ edges to gain, minus twice the preserved mass.

Step 2: YES-instances (l. 1626–1632). A solution maps the triples consecutively into the cycles; under the layout bijection π every edge goes to an edge, and only the endpoints of the paths gain one neighbour. Lean: `layout_adj_cycleUnion` gives $Q(N(v),N(\pi v)) = \deg(v)$ for $Q = P_\pi$, so $L_{P_\pi}(v,\pi v) = 2 - \deg(v) \le 1$ (each vertex has a neighbour as $a_i \ge 2$); `costOne_le` and `dOTOne_le` conclude.

Step 3: NO-instances, reduction to Eq. (44) (p. 52, l. 1633–1644). The paper: if every blown-up path went into one blown-up cycle (Eq. (44)), counting would give a solution; so some path is split, and Cases 1 and 2 show that some vertex has mismatch $\ge 2k$. Lean (contrapositive): suppose $C_1(Q) < 2$ for some $Q \in \mathrm{DS}$ (via `le_dOTOne`). Then by Step 1, for every $(v,w) \in \mathrm{supp}(Q)$, $\deg(v) < 2\,Q(N(v),N(w))$ (local step `hloc`).

Step 4: Case 2, a split cloud (l. 1653–1663). The paper: if $(v,j),(v,j')$ go to different cycles, the neighbourhoods of their images are disjoint, so $\ell + \ell' \le dk$ and one of them has mismatch $\ge 2k$. Lean (local step `hrow`): if $Q(v,w_1), Q(v,w_2) > 0$ with w_1, w_2 in different cycles, then $Q(N(v),N(w_1)) + Q(N(v),N(w_2)) \le \deg(v)$ (the neighbourhoods are disjoint and the rows of Q sum to 1), contradicting $\deg(v) < 2Q(N(v),N(w_i))$ for $i = 1,2$. Hence every row of Q lies in one cycle.

Step 5: Case 1, adjacent clouds in different cycles (l. 1645–1652). The paper: v of degree two whose neighbour w goes to a different cycle loses k and gains $\geq k$ adjacencies. Lean (local step `hconst`): if the neighbour v'' of v goes to another cycle than v, the mass of v'' does not contribute, so $Q(N(v), N(w)) \leq \deg(v) - 1$ and $\deg(v) < 2\deg(v) - 2$, i.e. $\deg(v) > 2$, contradicting `deg_pathUnion_le`. The argument applies to either endpoint of the edge; it only uses $\deg(v) \leq 2$.

Step 6: counting (after Eq. (44), l. 1639–1642). Every path of F_1 goes into a single cycle (`eq_of_pathUnion_adj`); as all cycles have A vertices, `isYes_of_isDS` produces a solution, contradicting the NO-assumption.

5 Assumptions and trusted base

No hypotheses: the results are unconditional and use only `propext`, `Classical.choice`, `Quot.sound` (checked by `scripts/CheckAxioms.lean`). The complexity-theoretic conclusion (polynomial construction, "unless P = NP") is not formalised; nor is Proposition 57 (p. 57, l. 1805–1808: NP membership and the 2^{nm}poly algorithm for δ_1^{OT}).

6 Notes on the formalisation

- **No blowups, no Theorem 1.** The paper bounds $d_1^\bullet$ via permutations of blowups and applies Theorem 1, which is formalised (`GsimOT.theorem1`). Following it in Lean would require, in addition, the combinatorial Cases 1 and 2 for permutations of k-blowups, the column-sum formula for those permutations, the passage between $\delta_{1\to1}^{\mathrm{OT}}$ (couplings) and d_1^{OT} (`deltaOTPQ_eq_dOT`, `dOT_one_one_eq_dOTOne`) and a limit argument. The direct proof has the same case distinction (Steps 4 and 5) on the matrix Q instead of on the cloud of a vertex, and it is at least as strong: applied to the doubly stochastic matrix Q_π induced by a blowup permutation π (as in the proof of Proposition 57), it would also give the paper's blowup bound (not formalised).
- **Hypotheses.** `dOTOne_le_of_isYes` assumes $a_i \geq 2$ for all i, `two_le_dOTOne_of_not_isYes` has no hypothesis on the a_i, and `dOTOne_shift_gap` holds for every instance.

7 How to check this result

From the repository root (after `export PATH=$HOME/.elan/bin:$PATH`):

- build the module:
  `lake build GsimOT.Theorem16_LocalEditHardness.Theorem16`
  (or, after a full build, `lake env lean GsimOT/Theorem16_LocalEditHardness/Theorem16.lean`);
- check the axioms: `bash build.sh` runs `scripts/CheckAxioms.lean`, which contains
  `#print axioms GsimOT.Hardness.ThreePartition.dOTOne_le_of_isYes`
  `#print axioms GsimOT.Hardness.ThreePartition.two_le_dOTOne_of_not_isYes`
  `#print axioms GsimOT.Hardness.ThreePartition.dOTOne_shift_gap`;
- the table `Theorem16_correspondence.pdf` in this folder lists every paper item with its Lean declaration and source line.